CA2: Cyber Attacks Analytics

VAST Challenge 2020: Honorable Mention for Streamlined Analysis Process

Luyu Cheng, Bairui Su, Yumeng Xue, Xiaoyu Liu, Yunhai Wang

Abstract—The mini-challenge 1 of VAST Challenge 2020 asks participants to find the responsible white hat groups for a fictitious Internet outage. For the given task, we designed and developed a visual analytics system – CA2: Cyber Attacks Analytics – to comparing and matching subgraphs of a very lary graph of anonymized profiles. Furthermore, we demonstrate an iterative workflow to find the responsible group with the help of our system.

Index Terms—Visual analytics, graph visualization, graph matching,

1 Introduction

Center for Global Cyber Strategy had collected and built a graph of profiles of the entire white hat community. The task is to identify the group that most closely resemble the given template group among 5 suspicious groups or all profiles. To achieve this, we present a visual analytics system – CA2: Cyber Attacks Analytics – and our solution to given tasks.

2 System Design

The challenge provides a very large graph, a template subgraph and 5 candidates subgraphs as input data. Each node represents one of following entities: persons, documents, demographics and countries. Each edge indicates one of following communication channels: authorship, sell, purchase, income, expenditure, and contact via phones or emails. Hence, our visual analytic system mainly focuses on the comparison and match over graphs.

2.1 Design Rationales

The design of the system is based on following rationales.

Comprehensive. The system should cover as many aspects of data as possible. In doing so, all possible aspects of data are exploited.

Hierarchical. All components in the system should be organized in a top-down hierarchy. Users can find the components they want to interact with by following the hierarchy.

Coordinated Multiple Views. Every component of the system links to its related components. Users' selections or modifications in one chart are supposed to reflected in related charts.

2.2 User Interface

Based on the rationales, we designed and developed the user interface (Fig. 1), which consists of three parts. The part (Fig. 1-1) on the left-most is the control panel. The part (Fig. 1-2) in the middle is the organization panel, which includes charts related to personal relationships. The part (Fig. 1-3) on the right is the personnel panel, containing charts showing personal data.

2.2.1 The Control Panel

The control panel has two functionalities: 1) adding, removing, and replacing subgraphs; and 2) enabling or disabling channels. Therefore, users are not limited to only provided template graph and candidate graphs because they could import any other subgraphs by uploading. Besides, users can toggle visibility of other channels in all components in order to focus on the channel they are interested in.

- *L. Cheng, B. Su, Y. Xue, X. Liu, and Y. Wang are with Shandong University, Qingdao. E-mail: {chengluyu, pearminipro, xym6336, lxysduer, cloudseawang}@gmail.com.*
- *Y. Wang is the corresponding author.*

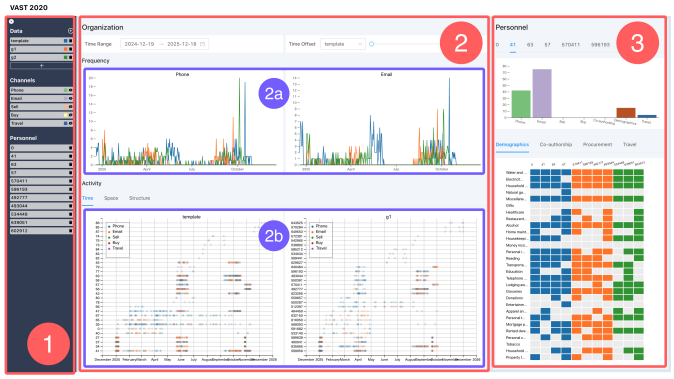


Fig. 1. An overview of the system, which consists of 3 major parts: control panel (1), organization panel (2), and personnel panel (3).

2.2.2 The Organization Panel

The organization panel provides views which display information about relationships between persons.

Users can set a time range and the time offset for each graph. This allows the user to efficiently match each subgraph and find the potential subgraph.

For organization visualization, the system first provides line charts (Fig. 1. (2a)) showing frequency of the overall activities of each organization. Users can get an overview of each subgraph in doing so. In the bottom (Fig. 1. (2b)), there are three views for visualizing activities, each view focuses on a particular aspect.

Time View. This view (Fig. 2. (a)) provides a temporal scatterplot of each person's activities in each organization. Here, each circle represents an activity record. A useful interaction is to move the cursor over a circle, connections between the activity's owner and other persons would be highlighted.

Space View. This view (Fig. 2. (b)) is a map to visualize the spatial distribution of each person's activities in each organization. Note that only the relative locations of individual countries are shown, not the exact latitude and longitude.

Structure View. This view (Fig. 2. (c)) shows the structure of each organization, and the people are connected by the activity they participate in. The user can click node to select the corresponding personnel. It is very helpful for the user to find the relationship between the template graph and the specific subgraph.

2.2.3 The Personnel Panel

The personnel panel (Fig. 3) focuses on information about selected individuals. The top contains several tabs, indicating selected persons respectively. Inside each tab, a bar chart shows the number of activities. Tabs inside the bottom (Fig. 3. b) corresponds channels selected in the control panel. Here, the heatmap depicts the frequency of all

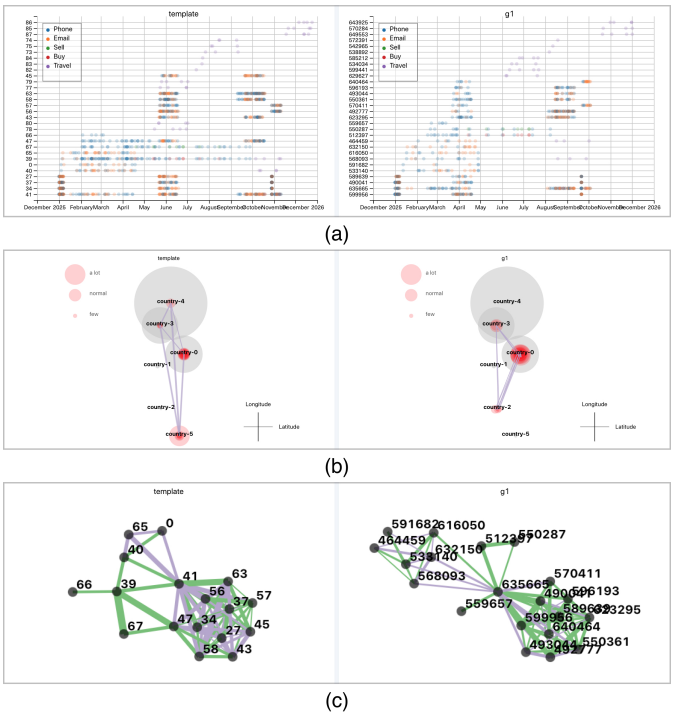


Fig. 2. Three activity visualizations in the organization panel: (a) time view, (b) space view, (c) structure view.

selected persons. By comparing between persons, users can determine the similarity between selected persons from different subgraphs and match them.

2.3 Implementation

The system is built around web-page technology. The charts and interactive graphs are created with D3.js [2] and the user interface is built with React [1]. The system is available online at `https://vast-2020.now.sh` and the source code is accessible at `https://github.com/pearmini/vast-2020`.

3 Large Graph Analysis

Based on the system we presented, we accomplished the task of comparing and matching the template graph with provided candidate graphs. Next, we will discuss how to employ the system to find another potential subgraph among the large graph provided.

There are over 100 million edges and over 200 thousand vertices in each template subgraph, hence we need to filter unrelated edges and vertices by matching them. For the sake of retrieval efficiency, we use a graph database to store and search the graph.

In order to find the matched nodes in the large graph, we analyzed the template graph by our visual analysis system. From the system, we conclude that only 2 people in the template graph are linked by procurement edges, and only one item node is related to the 2 people. Hence, we use the characteristics of the edges between these three nodes as conditions to find matched nodes in the large graph. Here, we propose the following iterative workflow that finds new nodes from matched nodes. By reproducing these steps, users could find subgraphs akin to the template graph from the large graph.

Step 1. Create a set S and put the matched nodes from the template into S; the remaining unmatched nodes are put into another set T. The template graph $G_T = \langle N_T, E_T \rangle$, while the large graph $G_L = \langle N_L, E_L \rangle$.

Step 2. Establish a map M for mapping the nodes in N_T and their corresponding nodes in N_L.

Step 3. For each unmatched node $T_i \in T$ who has $A \in Adjacent(T_i) \in S$, get the node set $\beta = Adjacent(M(A))$. For each $\beta_j \in \beta$, compare the similarity between $Edges\langle A, T_i \rangle$ and $Edges\langle M(A), \beta_j \rangle$, and return the most similar node peer $\langle T_i, \beta_j \rangle$.

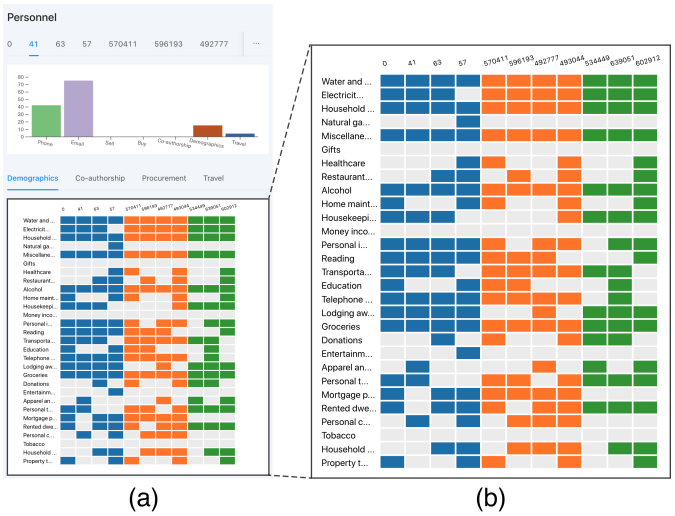


Fig. 3. The personnel panel (a) of the system. Users can inspect selected persons' information in the top and comparing selected persons in terms of activities in the bottom (b).

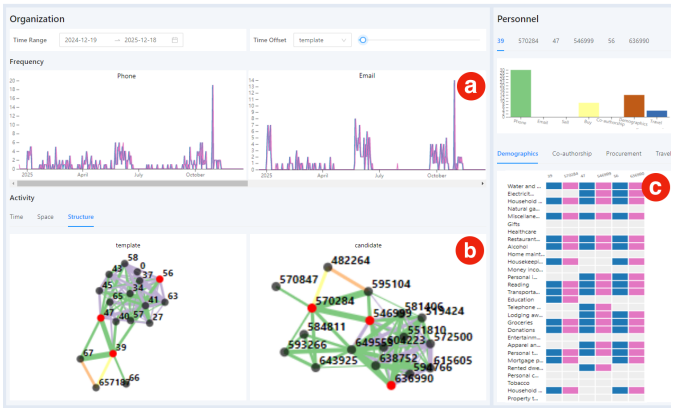


Fig. 4. The result of large graph matching. The candidate graph and the template graph shows a high similarity in the frequency chart (a), the graph structure (b), and the heat map (c).

Step 4. For each node peer $\langle T_i, \beta_j \rangle$ returned by **Step 3**, visualize the graph $\langle S \cup T_i, Edges\langle S, S \rangle \cup Edges\langle S, T_i \rangle \rangle$ and the graph $\langle M(S) \cup \beta_i, Edges\langle M(S), M(S) \rangle \cup Edges\langle M(S), \beta_j \rangle \rangle$. Then determined by the user whether it matches or not. If matched, update $M(T_i) = \beta_j$ and add T_i into S while deleting T_i from T.

Step 5. If set T is empty, then exit; otherwise, execute **Step 3** again.

As depicted in Fig. 4, the obtained subgraph is almost identical to the template graph. We have put the subgraph on the online demo. The subgraph can be visited by adding a new subgraph called "candidate" in the control panel.

4 Conclusion

In this report, we present – CA2: Cyber Attacks Analytics – a visual analytics system that is capable of comparing and matching the template subgraph with candidate graphs in terms of many aspects. The tool is suitable for rapidly locating the responsible white hat group and thus serves needs of VAST Challenge 2020 Mini-Challenge 1.

References

[1] React – a javascript library for building user interfaces `https://reactjs.org/`.

[2] M. Bostock, V. Ogievetsky, and J. Heer. D³ data-driven documents. *IEEE Transactions on Visualization and Computer Graphics*, 17(12):2301–2309, 2011.